Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Pre-Tutorial Assignment 4: Solutions

---

1. Consider the electric circuit shown in Figure 1. The circuit elements have the following numerical values – $R_1 = 20\Omega$, $R_2 = 10\Omega$, $R_3 = 30\Omega$, $I_1 = 4A$, $V_1 = 60V$, $I_d = k \times i = 0.4 \times i$. Evaluate the voltage V_{R3} across the resistor R_3 using the *Superposition Theorem*. Report the numerical values of $V_{R3}(I_1)$ (when only I_1 is active) and $V_{R3}(V_1)$ (when only V_1 is active).

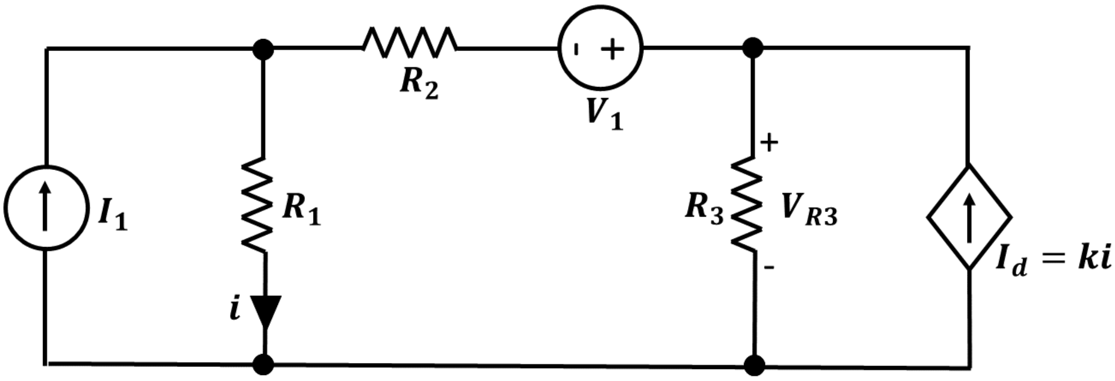


Figure 1: Electric circuit with two independent sources and one dependent source.

Solution

The voltage V_{R3} across the resistor R_3 can be obtained by adding the voltages measured in two individual circuits, each having one active independent source. Thus, by using the Superposition theorem, $V_{R3} = V_{R3}\,(I_1) + V_{R3}\,(V_1)$

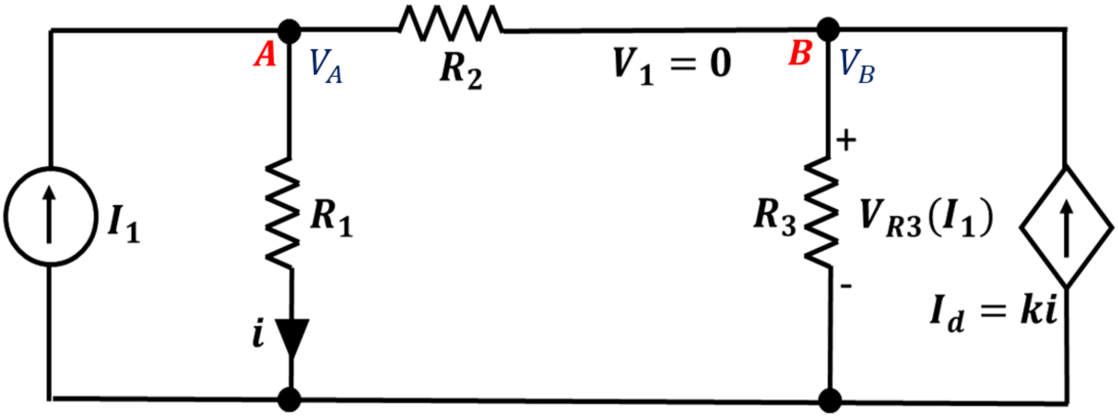


Figure 2: The independent voltage source is deactivated and only the current source I_1 is active.

Figure 2 shows the circuit with only source I_1 active. Application of KCL at Node **A** gives us

$$
\begin{aligned}
I_1 - G_1 V_A - G_2(V_A - V_B) &= 0 \qquad (1)\\
\Rightarrow \quad (G_1 + G_2)V_A - G_2 V_B &= I_1 \qquad (2)
\end{aligned}
$$

Application of KCL at Node **B** gives us

$$\begin{aligned} G_2(V_A - V_B) - G_3V_B + I_d &= 0 \quad (3) \\ \Rightarrow \quad -(kG_1 + G_2)V_A + (G_2 + G_3)V_B &= 0 \end{aligned}$$

(4)

Using the numerical values and by solving equations 2 and 4, we obtain $V_{R3}(I_1) = V_B = 60V$

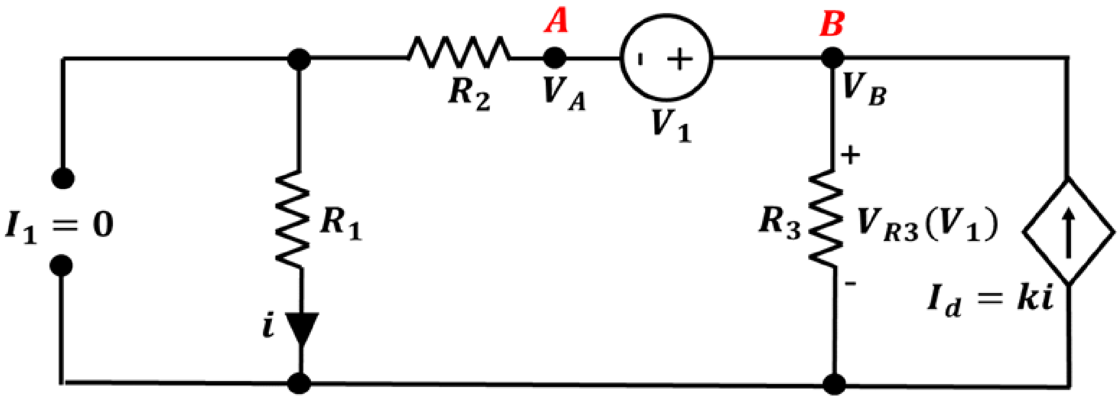


Figure 3: The independent current source is deactivated and only the voltage source V_1 is active.

Figure ?? shows the circuit with only source V_1 active. Application of KCL at Node **B** and an inspection of the circuit gives us the following equations.

$$\begin{aligned} -i - G3V_B + I_d &= 0 \quad (5) \\ i &= \frac{V_A}{R_1 + R_2} \quad (6) \\ V_B - V_A &= V_1 \quad (7) \\ & \quad (8) \end{aligned}$$

Using the above equations, we obtain the following

$$\begin{aligned} \frac{(k-1)(V_B - V_1)}{R_1 + R_2} &= G_3V_B \quad (9) \\ \Rightarrow V_B &= \frac{\alpha V_1}{\alpha - G_3} \quad (10) \\ & \quad (11) \end{aligned}$$

where $\alpha = \frac{k-1}{R_1+R_2}$. Using the numerical values, we obtain $V_{R3}(V_1) = 22.5V$
Thus, using the Superposition theorem, we obtain $V_{R3} = V_{R3}(I_1) + V_{R3}(V_1) = 82.5V$

2. Consider the Boolean function

$$F(A, B, C) = \Sigma(0, 1, 2, 5, 6, 7).$$

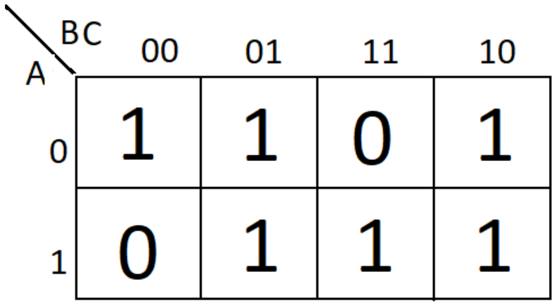
(a) Draw the Karnaugh map for the function F.

(b) Find all the Boolean expressions for F in the minimal SOP form.

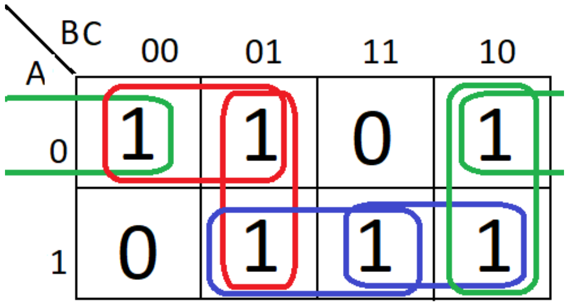
(c) Find all the Boolean expressions for F in the minimal POS form.

Solution

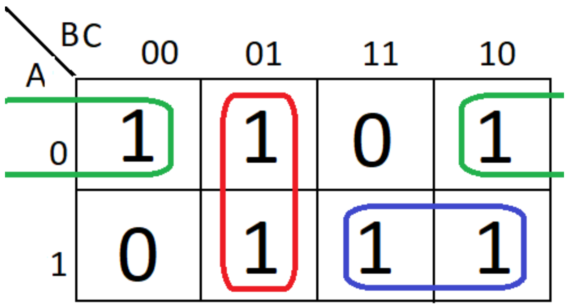
(a) The K-map for the function F is



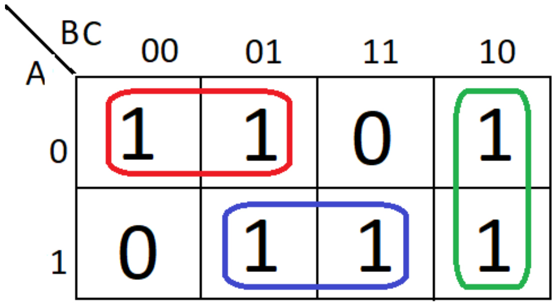
(b) In the K-map above, each 1 can be covered by two groupings of 1's as shown below. Thus the



minimal SOP expressions may not be unique. The minimal SOP expressions can be found by considering the following coverings:

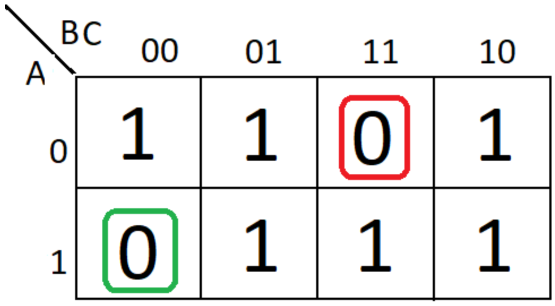


$$F(A, B, C) = \overline{A}.\overline{B} + A.C + B.\overline{C}$$



$$F(A,B,C) = \overline{A}.\overline{B} + A.C + B.\overline{C}$$

(c) In the K-map above, we can consider the coverings of 0's as shown below. The unique minimal



POS expression is

$$(\overline{A} + B + C).(A + \overline{B} + \overline{C})$$

.